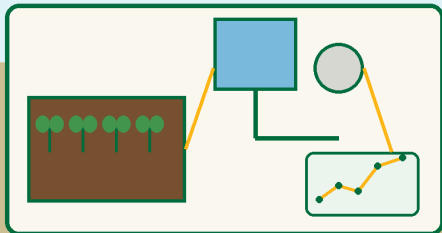


ENGINEERING PROBLEM

PROBLEM



PROBLEM

ENGINEERING PROBLEM

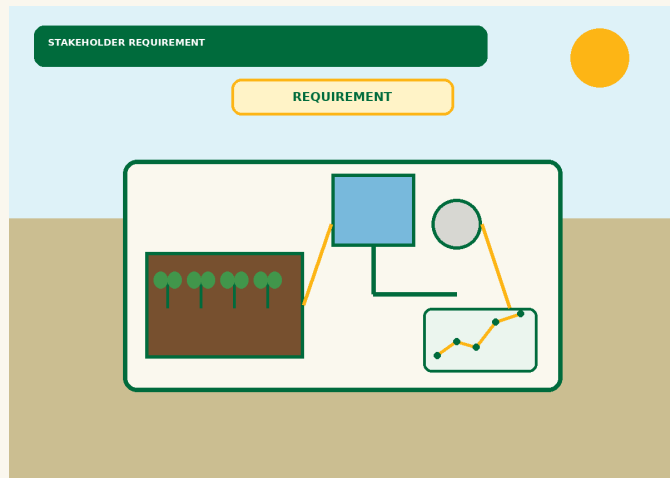
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

An engineering problem statement defines an unmet need, system context, affected users, constraints, and measurable performance gap.

ANALYSIS / DESIGN TASK

Write a concise problem statement for a garden system.



REQUIREMENT

STAKEHOLDER REQUIREMENT

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

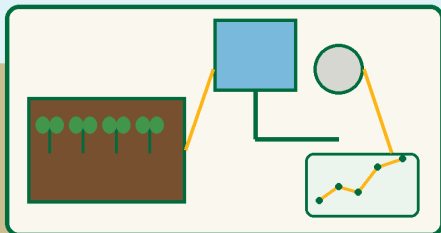
A stakeholder requirement describes a needed function, performance level, usability condition, or outcome from the perspective of a user or decision-maker.

ANALYSIS / DESIGN TASK

Elicit and rewrite one requirement measurably.

DESIGN REQUIREMENT

DESIGN REQ



DESIGN REQ

DESIGN REQUIREMENT

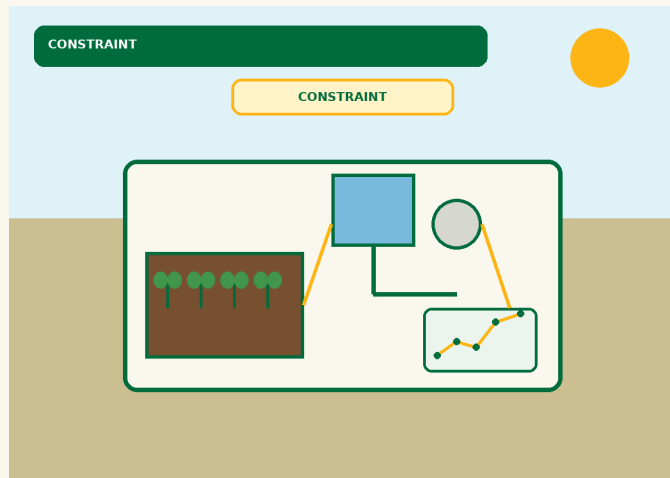
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A design requirement translates needs into testable technical criteria that a proposed solution must satisfy.

ANALYSIS / DESIGN TASK

Create three measurable requirements.



CONSTRAINT

CONSTRAINT

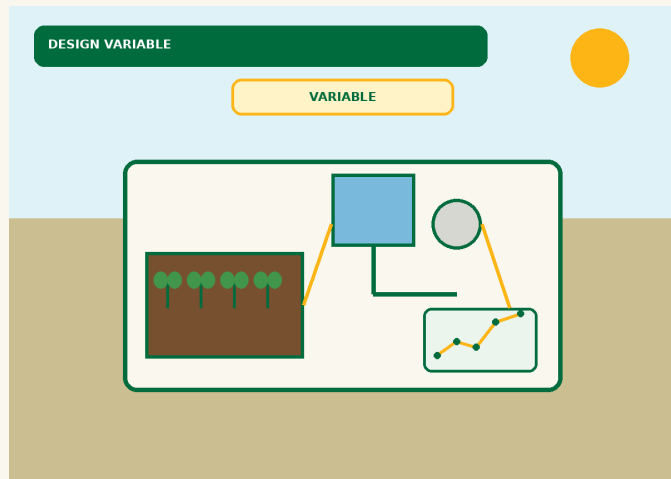
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A design constraint is a non-negotiable limit involving budget, safety, space, resources, regulations, materials, or time.

ANALYSIS / DESIGN TASK

List and rank project constraints.



VARIABLE

DESIGN VARIABLE

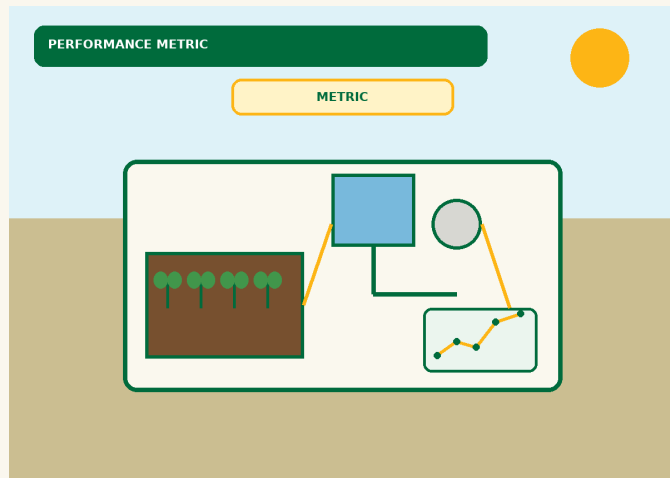
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A design variable is a controllable parameter that can be changed during design, such as size, spacing, flow, storage, or operating threshold.

ANALYSIS / DESIGN TASK

Identify four design variables.



METRIC

PERFORMANCE METRIC

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

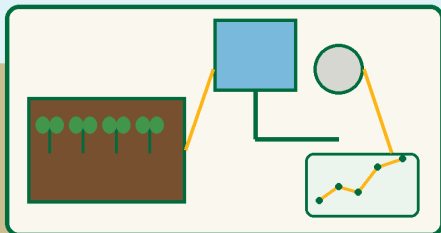
A performance metric quantifies how well a design meets an objective or requirement.

ANALYSIS / DESIGN TASK

Define metrics with units.

DESIGN ALTERNATIVE

ALTERNATIVE



ALTERNATIVE

DESIGN ALTERNATIVE

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

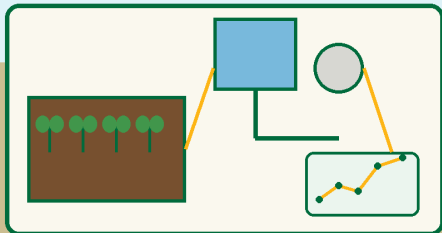
A design alternative is a distinct feasible configuration or strategy developed for comparison.

ANALYSIS / DESIGN TASK

Generate three alternatives.

CONCEPT SCREENING

SCREEN



SCREEN

CONCEPT SCREENING

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Concept screening eliminates alternatives that fail essential requirements or constraints before detailed comparison.

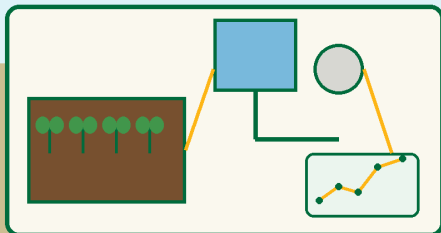
ANALYSIS / DESIGN TASK

Screen alternatives against mandatory criteria.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

DECISION MATRIX

MATRIX



MATRIX

DECISION MATRIX

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A decision matrix compares alternatives across weighted criteria using an explicit scoring method.

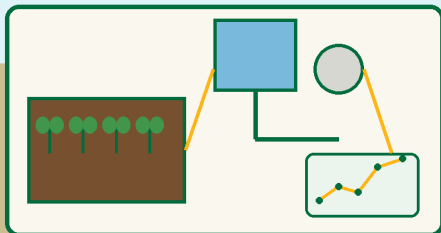
ANALYSIS / DESIGN TASK

Build and calculate a simple weighted matrix.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

TRADE-OFF ANALYSIS

TRADE-OFF



TRADE-OFF

TRADE-OFF ANALYSIS

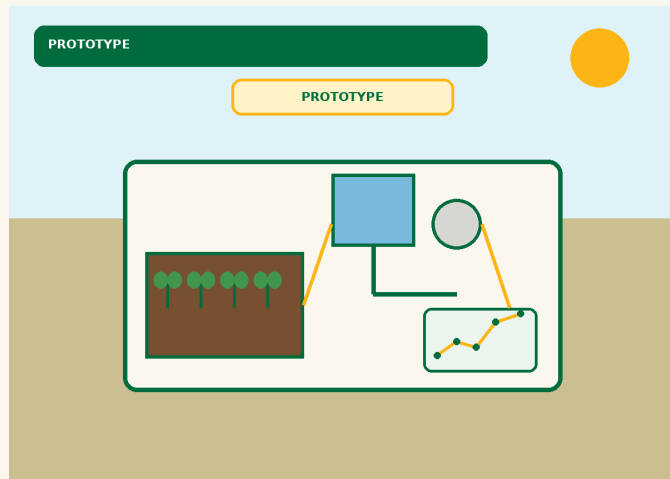
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Trade-off analysis examines competing objectives where improvement in one criterion may reduce another.

ANALYSIS / DESIGN TASK

Plot or describe a two-objective trade-off.



PROTOTYPE

PROTOTYPE

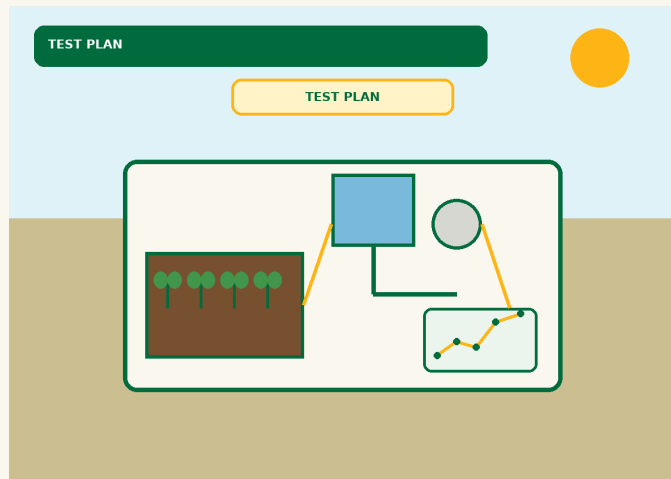
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A prototype is a simplified physical, computational, or functional representation used to test important design assumptions.

ANALYSIS / DESIGN TASK

Define what a prototype must demonstrate.



TEST PLAN

TEST PLAN

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

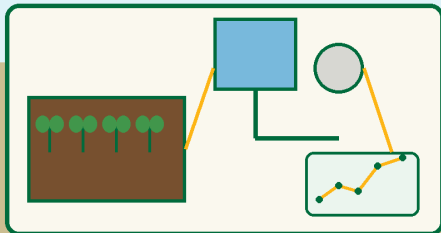
A test plan specifies performance questions, procedures, instruments, acceptance criteria, repetitions, and data to collect.

ANALYSIS / DESIGN TASK

Draft a test plan for one requirement.

ACCEPTANCE CRITERION

ACCEPTANCE



ACCEPTANCE

ACCEPTANCE CRITERION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

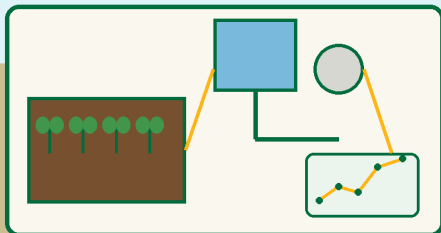
An acceptance criterion is a measurable threshold that determines whether a tested requirement has been satisfied.

ANALYSIS / DESIGN TASK

Write one pass/fail criterion.

DESIGN ITERATION

ITERATE



ITERATE

DESIGN ITERATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Design iteration uses test evidence to modify a solution and retest performance.

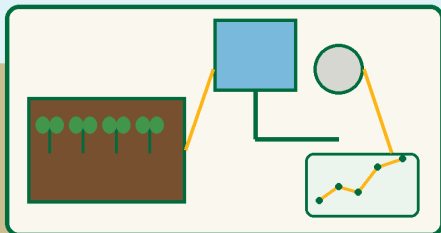
ANALYSIS / DESIGN TASK

Document one evidence-driven revision.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

TECHNICAL RISK

RISK



RISK

TECHNICAL RISK

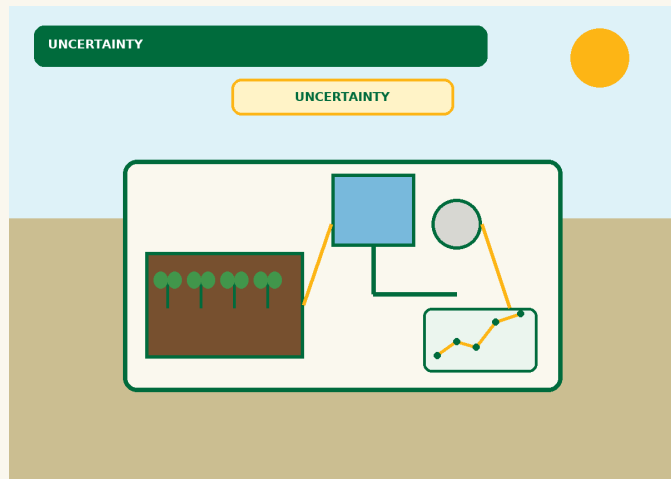
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Technical risk combines the likelihood and consequence of failure or underperformance in a design element.

ANALYSIS / DESIGN TASK

Create a simple risk register.



UNCERTAINTY

UNCERTAINTY

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

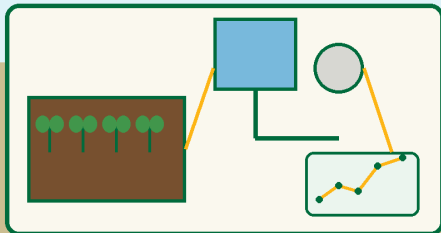
Design uncertainty arises from incomplete knowledge of loads, environment, users, component performance, models, and future conditions.

ANALYSIS / DESIGN TASK

Identify and rank uncertainty sources.

LIFE-CYCLE THINKING

LIFE CYCLE



LIFE CYCLE

LIFE-CYCLE THINKING

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Life-cycle thinking considers material acquisition, construction, operation, maintenance, repair, and end-of-life implications.

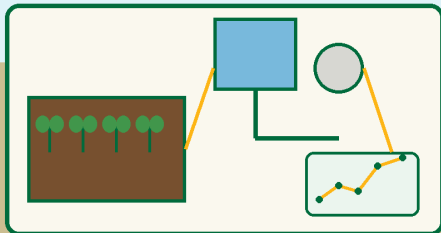
ANALYSIS / DESIGN TASK

Map stages for one garden technology.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

TECHNO-ECONOMIC METRIC

ECONOMICS



ECONOMICS

TECHNO-ECONOMIC METRIC

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Techno-economic metrics connect technical performance with costs, benefits, lifetime, and operating assumptions.

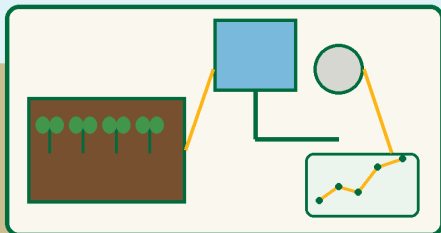
ANALYSIS / DESIGN TASK

Calculate a simple annualized or unit-cost comparison.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

DESIGN DOCUMENTATION

DOCUMENT



DOCUMENT

DESIGN DOCUMENTATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

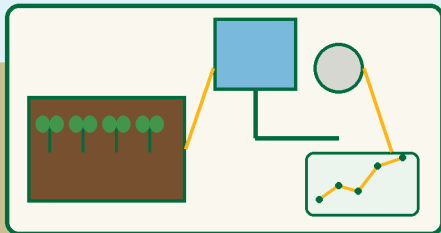
Engineering documentation preserves requirements, assumptions, drawings, calculations, data, decisions, revisions, and test results.

ANALYSIS / DESIGN TASK

Audit a project record for traceability.

ENGINEERING DESIGN REVIEW

REVIEW



REVIEW

ENGINEERING DESIGN REVIEW

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A design review determines whether evidence supports progression, revision, or rejection of a proposed solution.

ANALYSIS / DESIGN TASK

Present a recommendation supported by requirements, tests, risks, and costs.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE